

ARCA ONLINE

System Documentation

Web-Based Inventory & Sales Management System

PHP · MySQL · Rest

Version 1.0 · 2025

1. Introduction

1.1 What is Arca Online?

Arca Online is a web-based management system designed to handle inventory, sales, and user management for small businesses or organizations. Built using PHP and MySQL, it provides a streamlined interface for tracking products, processing transactions, generating receipts, and managing user access.

The system was developed by *Renier Tambogon* and his friends progressively as part of a BSIT learning journey, evolving from simple PHP pages into a structured, multi-layered application. Despite its educational roots, Arca Online reflects real-world development practices including frontend and backend separation, session-based authentication, role management, and relational database design.

1.2 Project Overview

Property	Details
System Name	Arca Online
Type	Web-Based Management System
Technology	PHP, MySQL, HTML, CSS, JavaScript
Purpose	Inventory tracking, sales management, user authentication
Development Stage	Active — evolving toward MVC architecture

1.3 Core Features

- Inventory Management — add, update, and delete items with stock and price tracking
- Sales Tracking — record transactions and link them to inventory
- Receipt Generation — automatically produce receipts per sale
- User Authentication — login, signup, logout with session handling
- Responsive Dashboard — mobile-friendly UI with sidebar navigation

2. System Architecture

2.1 Architecture Overview

Arca Online follows a simplified layered architecture. It separates concerns across three distinct layers: the frontend (presentation), the backend (logic), and the database (data persistence). While not a full MVC implementation yet, the structure mirrors those principles.

Request Flow

```
User (Browser)
|
v
Frontend Layer  (inventory.php, sales.php, main.php)
|
v
Backend Layer   (add.php, update.php, delete.php, login.php)
|
v
Database Layer  (MySQL: items, sales, users, receipt)
```

2.2 Folder Structure

```
arca-online.
|
|  index.php
|  README.md
|  structure.txt
|
├── client
|   |
|   |  index.php
|   |
|   ├── admin
|   |   |
|   |   |  admin.php
|   |
|   ├── assets
|   |   |
|   |   |  ├── css
|   |   |   |
|   |   |   |  counter.css
|   |   |   |  dashboard.css
|   |   |   |  index.css
|   |   |   |  inventory.css
|   |   |   |  sales.css
|   |   |   |  style.css
|   |   |
|   |   ├── img
|   |   |   |
|   |   |   |  arca.png
|   |   |   |  logo.png
|   |   |
|   |   └── js
|   |       |
|   |       |  login.js
|   |       |  register.js
|   |
|   └── auth
|       |
|       |  login.html
|       |  register.html
|   └── pages
|       |
|       |  counter.php
|       |  dashboard.php
|       |  inventory.php
|       |  sales.php
```

```
├── config
│   └── connect.php
├── database
│   ├── inventorydb-schema.png
│   └── inventorydb.sql
├── server
│   ├── control.php
│   ├── auth
│   │   ├── authenticate.php
│   │   ├── login.php
│   │   ├── logout.php
│   │   └── signup.php
│   ├── counter
│   │   ├── cancel.php
│   │   ├── decrease.php
│   │   ├── punch.php
│   │   ├── push.php
│   │   └── receipt.php
│   └── inventory
│       ├── add.php
│       ├── delete.php
│       ├── fetch.php
│       └── update.php
```

2.3 Frontend Layer

The frontend layer handles all user-facing pages and interactions. It communicates with the backend through form submissions and AJAX/Fetch calls.

- Displays inventory lists, sales records, and dashboards
- Renders forms for adding and editing data
- Handles responsive layout and sidebar navigation
- Technologies: HTML, CSS, JavaScript, PHP templates

2.4 Backend Layer

The backend layer acts as the processing core of the system. It receives requests from the frontend, validates input, runs business logic, and interacts with the database.

- Handles all CRUD operations (Create, Read, Update, Delete)
- Processes authentication and session management
- Manages inventory updates and sales recording
- Generates receipt data
- Technologies: PHP, MySQLi

2.5 Database Layer

The database layer stores all persistent application data using a relational MySQL structure.

Property	Details
users	Stores user accounts, hashed passwords, and role assignments
items	Stores product information including name, stock quantity, and price
sales	Stores transaction records linked to items and users
receipt	Stores generated receipt data per completed transaction

3. Authentication System

3.1 Overview

Arca Online uses a session-based authentication system. When a user logs in with valid credentials, a PHP session is created to maintain their authenticated state across protected pages.

3.2 Authentication Flow

```
User submits login form
      |
      v
login.php validates input
      |
      v
Database checks credentials
      |
  ----+----
  |      |
Valid   Invalid
  |      |
  v      v
Session Error returned
created  to user
  |
  v
Redirect to dashboard
```

3.3 Key Concepts

- `session_start()` — initiates or resumes a user session on every protected page

- Credential validation — credentials submitted via POST are checked against the database
- Session variables — store authenticated user data (user ID, role) during the session
- Logout — destroys the session completely via `session_destroy()`

4. Security

4.1 Security Overview

Arca Online implements several security layers appropriate for its development stage. The system protects authenticated routes, validates user input, and controls access based on user roles.

4.2 Security Measures

Session-Based Authentication

All protected pages check for an active session before rendering content. If no valid session exists, the user is immediately redirected to the login page.

Role-Based Access Control (RBAC)

User roles are stored in the database and checked server-side. Different roles such as admin vs. regular user are granted access to different pages and actions. Role information is stored in the session after login and validated on each request.

Input Validation

All form data sent to the backend is validated before any database operation is performed. This reduces the risk of invalid or malformed data being inserted into the system.

Password Handling

User passwords are not stored in plain text. The system uses PHP hashing functions to store passwords securely and verifies them during login without exposing the original password.

4.3 Security Summary

Property	Details
Session Auth	PHP sessions protect all authenticated routes from unauthorized access
RBAC	User roles control access to admin features and sensitive operations

Input Validation	Backend validates all POST data before running any database query
Password Hashing	Passwords are hashed before storage and never stored in plain text
Logout	Sessions are fully destroyed on logout to prevent session reuse

4.4 Recommended Future Security Improvements

- Prepared statements to prevent SQL injection attacks
- CSRF tokens on all forms to prevent cross-site request forgery
- HTTPS enforcement in all deployment environments
- Rate limiting on login attempts to prevent brute force attacks
- More granular permission management per user role

5. Development Journey

Arca Online evolved in stages, each building on the previous. This progression mirrors how real-world systems grow from simple beginnings into structured applications.

Property	Details
Stage 1	Basic PHP pages — CRUD operations, form handling, database connections
Stage 2	Authentication — login, signup, logout, sessions, user validation
Stage 3	Inventory & Sales — item management, sales tracking, receipt generation
Stage 4	Frontend/Backend Separation — organized folder structure, better maintainability
Stage 5	UI/UX Improvements — responsive design, sidebar nav, black and gold theme
Stage 6	System Thinking — architecture awareness, MVC exploration, scalability planning

6. Technologies Used

Backend

- PHP — server-side scripting and business logic
- MySQL — relational database management

- MySQLi — PHP extension for database connections
- PHP Sessions — authentication and state management

Frontend

- HTML — page structure and markup
- CSS — styling, layout, and responsive design
- JavaScript — client-side interactions and dynamic behavior

Development Tools

- Git & GitHub — version control and repository management
- VS Code — primary code editor
- XAMPP — local development server (Apache + MySQL + PHP)

Deployment & Hosting

- InfinityFree / Hostinger — web hosting exploration
- Git workflows — pull requests and deployment debugging

7. Future Roadmap

Arca Online is actively evolving. The following improvements are planned for future development:

- MVC Architecture — cleanly separate Model, View, and Controller responsibilities
- Prepared Statements — replace raw queries with parameterized queries for SQL injection prevention
- REST API Layer — expose backend functionality through clean API endpoints
- Reusable Components — shared UI components for consistent design across pages
- Improved Database Design — normalized schema with proper foreign key constraints
- CSRF Protection — add tokens to all forms
- Automated Deployment — Git-based deployment pipeline for production hosting